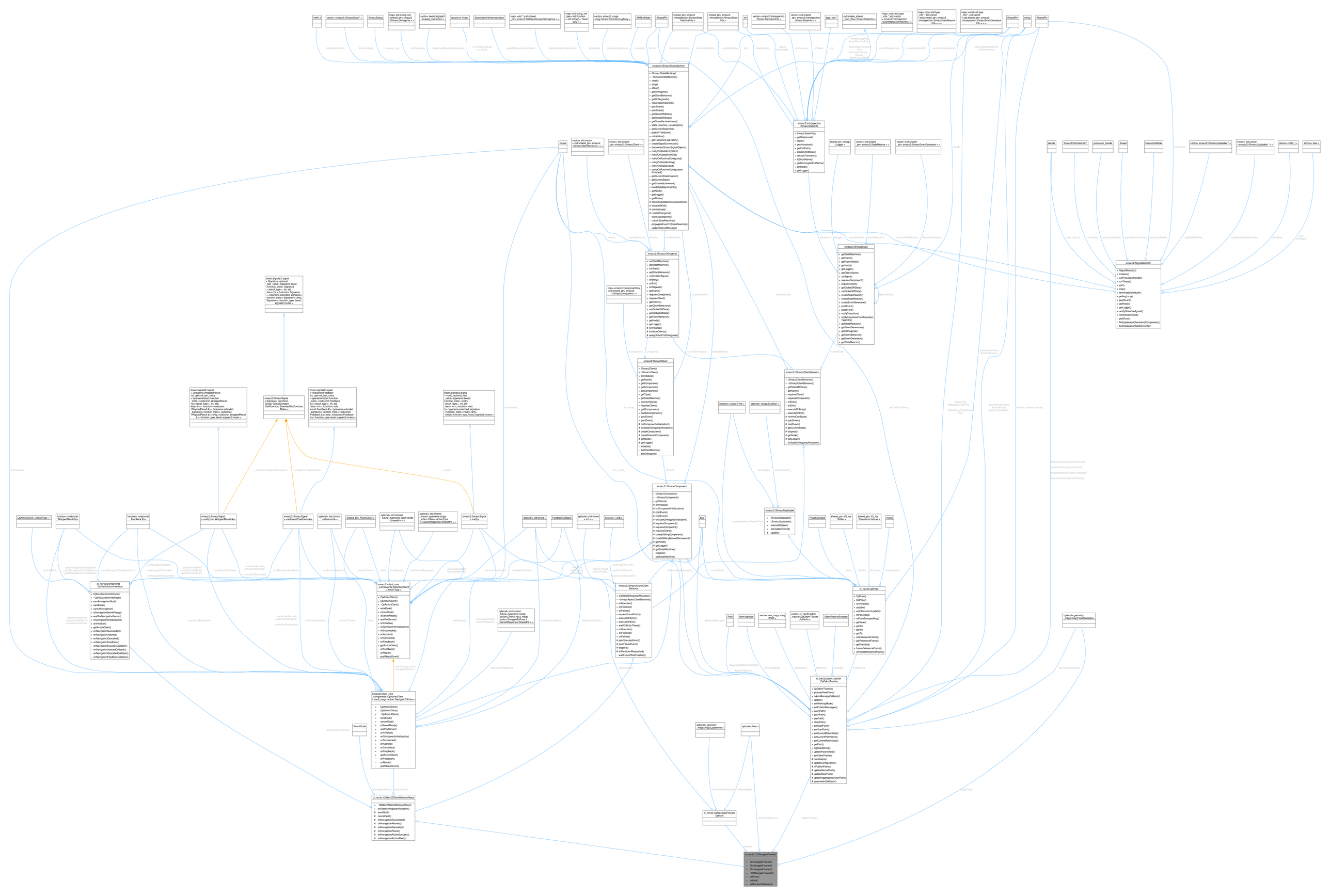

int64_t
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
vector< boost::signals2::scoped_connection >
recursive_mutex
StateMachineInternalAction
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
SharedPtr
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
shared_ptr< smacc2::introspection::SmaccStateInfo >
int
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
SharedPtr
string
SharedPtr
-stateSeqCounter_
#currentState_
#status_msg_
#orthogonals_
#longLivedSignalConnections_
-eventQueueMutex_ -m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
#timer_
#stateMachineInfo_
#currentStateInfo_
+stateMachine_
#nh_
+parentState_
+depth_ +stateIndex_
+transitions_
+children_
+tid_
#eventsLogPub_ #stateMachinePub_ #stateMachineStatusPub_ #transitionHistoryService_ #transitionLogPub_
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+demangledStateName +fullStateName
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExitting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
mutex
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
vector< std::shared_ptr< smacc2::ISmaccClient > >
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
#node_
-statusPub_
double
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
atomic< int64_t >
atomic< bool >
-mutex_
-clientBehaviors_
#clients_
-stateMachine_
#stateInfo_
#logger_
-signalDetector_
#stateReactors_
-smaccStateMachine_
#eventGenerators_
-loop_rate_hz
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
-lastState_
-end_ -initialized_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
-stateMachine_
-stateMachine_
#components_
-orthogonal_
-currentOrthogonal
#stateMachine_
-currentState
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
boost::signals2::signal< void(const WrappedResult &), optional_last_value< typename boost::function_traits< void(const WrappedResult &)>::result_type >, int, std::less< int >, function< void(const WrappedResult &)>, typename extended_signature< function_traits< void(const WrappedResult &)>::arity, void(const WrappedResult &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(const Feedback &), optional_last_value< typename boost::function_traits< void(const Feedback &)>::result_type >, int, std::less< int >, function< void (const Feedback &)>, typename extended_signature< function_traits< void(const Feedback &)>::arity, void(const Feedback &)>::function_type, boost::signals2::mutex >
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void ()>, typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
optional< rclcpp::Time >
optional< rclcpp::Duration >
-actionMutex_
-m_mutex_
#robotPoseTimer_
#odomSub_ #robotBasePathPub_ #robotBasePathStackedPub_
-poseFrameName_ -referenceFrame_
#currentPathName_ #odomFrame_ #odomTopicName_
#param_callback_handle_
#clearAngularDistanceThreshold_
#clearPointDistanceThreshold_
#recordAngularDistanceThreshold_
#recordPointDistanceThreshold_
< void(const WrappedResult &)>
< void(const Feedback &)>
< void()>
#m_mutex_
#owner_
-lastUpdate_
-periodDuration_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
+frozenReferenceFrameTime
CpActionClient< ActionType > *
function< void(const WrappedResult &)>
function< void(const Feedback &)>
smacc2::SmaccSignal< void(const WrappedResult &)>
smacc2::SmaccSignal< void(const Feedback &)>
optional< std::chrono::milliseconds >
shared_ptr< ActionClient >
optional< std::shared_future< typename GoalHandle::SharedPtr > >
optional< std::shared_future< typename rclcpp_action::Client< ActionType >::CancelResponse::SharedPtr > >
smacc2::SmaccSignal< void()>
optional< std::string >
FeedbackCallback
optional< std::future< int > >
function< void()>
bool
PoseStamped
shared_ptr< tf2_ros::Buffer >
shared_ptr< tf2_ros::TransformListener >
mutex
-actionClient_
+postNavigationAbortedEvent +postNavigationCancelledEvent +postNavigationSuccessEvent
+postNavigationFeedbackEvent
+onNavigationAborted_ +onNavigationCancelled_ +onNavigationSucceeded_
+postAbortedEvent +postCancelledEvent +postSuccessEvent
+postFeedbackEvent
+onActionAborted_ +onActionCancelled_ +onActionSucceeded_
+onNavigationFeedback_
+onActionFeedback_
+serverTimeout
-client_
-lastRequest_
-lastCancelResponse_
+onGoalAccepted_ +onGoalRejected_
+actionServerName
-feedbackCallback_
-onFailure_ -onFinished_ -onSuccess_
-onEntryThread_ -onExitThread_
-postFailureEventFn_ -postFinishEventFn_ -postSuccessEventFn_
-isShutdownRequested_
+isInitialized
-pose_
-tfBuffer_
-tfListener_
-listenerMutex_
cl_nav2z::components::CpNav2ActionInterface
+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()
smacc2::client_core_components::CpActionClient< ActionType >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
cl_nav2z::CpPose
+ CpPose()
+ CpPose()
+ onInitialize()
+ update()
+ waitTransformUpdate()
+ toPoseMsg()
+ toPoseStampedMsg()
+ getYaw()
+ getX()
+ getY()
+ getZ()
+ setReferenceFrame()
+ getReferenceFrame()
+ getFrameId()
+ freezeReferenceFrame()
+ unfreezeReferenceFrame()
optional< std::shared_future< typename rclcpp_action::Client< nav2_msgs::action::NavigateToPose >::CancelResponse::SharedPtr > >
-lastCancelResponse_
-lastRequest_
+onGoalAccepted_ +onGoalRejected_
+actionServerName
-feedbackCallback_
+goalChecker_
#publishMessages
Path
WorkingMode
vector< nav_msgs::msg::Path >
vector< cl_nav2z::odom_tracker::CpOdomTracker::PathInfo >
OdomTrackerStrategy
optional< geometry_msgs::msg::PoseStamped >
#aggregatedStackPathMsg_ #baseTrajectory_
#workingMode_
#pathStack_
#pathInfos_
#strategy_
#robotPose_
#currentMotionGoal_
< nav2_msgs::action::NavigateToPose >
smacc2::client_core_components::CpActionClient< nav2_msgs::action::NavigateToPose >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
ResultCode
#nav2ActionInterface_
cl_nav2z::odom_tracker::CpOdomTracker
+ CpOdomTracker()
+ processNewPose()
+ odomMessageCallback()
+ update()
+ setWorkingMode()
+ setPublishMessages()
+ pushPath()
+ pushPath()
+ popPath()
+ clearPath()
+ setStartPoint()
+ setStartPoint()
+ setCurrentMotionGoal()
+ setCurrentPathName()
+ getCurrentMotionGoal()
+ getPath()
+ logStateString()
+ updateParameters()
+ setOdomFrame()
# onInitialize()
# updateConfiguration()
# rtPublishPaths()
# updateRecordPath()
# updateClearPath()
# updateAggregatedStackPath()
# parametersCallback()
optional< geometry_msgs::msg::Quaternion >
optional< float >
+forceInitialOrientation
+forwardSpeed
#navigationResult_
#actionClient_
cl_nav2z::CbNav2ZClientBehaviorBase
+ ~CbNav2ZClientBehaviorBase()
+ onStateOrthogonalAllocation()
# sendGoal()
# cancelGoal()
# onNavigationSucceeded()
# onNavigationAborted()
# onNavigationCancelled()
# onNavigationResult()
# onNavigationActionSuccess()
# onNavigationActionAbort()
cl_nav2z::CbNavigateForwardOptions
#forwardDistance_
#odomTracker_
#goalPose_
+options
cl_nav2z::CbNavigateForward
+ CbNavigateForward()
+ CbNavigateForward()
+ CbNavigateForward()
+ ~CbNavigateForward()
+ onEntry()
+ onExit()
+ setForwardDistance()